Distributions Handout

AMC 10 Preparation

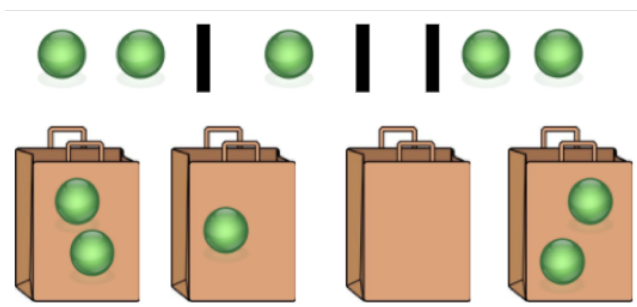
THEORY

Many counting problems involve some incarnation of placing n balls in k distinguishable bins or urns. This is commonly referred to as the *ball and urn* problem.

This is the equivalent of finding the number of tuples of non-negative integers $(x_1, x_2, ..., x_k)$ that satisfy:

$$\sum_{i=1}^{k} x_i = x_1 + x_2 + \cdots + x_k = n.$$

We establish a *bijection* (covered more thoroughly in Lecture 6) between the number of ball distributions and the number of ways to place dividers among a row of balls. To partition n balls into k bins, we need $k-1$ dividers, as shown below for $k = 4$ and $n = 5$.



We note that for any arrangement of these n balls and $k-1$ dividers, we form a *unique* distribution of the balls into bins. There are $n+k-1$ total slots, and thus

$$\boxed{\binom{n+k-1}{k-1}}$$

ways to place the $k-1$ dividers. Since the n balls must fall into the remaining places, this formula equals the total number of possible distributions. We can also consider the case where we must place at least one ball inside each bin. This can be accomplished by realizing by first placing the mandatory k balls, and noting that we have $n-k$ balls remaining that can go anywhere. We must solve the following for $x_1^* = x_i - 1 \geq 0$.

$$\begin{aligned}(x_1 - 1) + (x_2 - 1) + \cdots + (x_k - 1) &= n - k \\ x_1^* + x_2^* + \cdots + x_k^* &= n - k\end{aligned}$$

Thus, we have $n-k$ balls and $k-1$ dividers, and thus $n-k+(k-1)=n-1$ total slots. This means that there are

$$\boxed{\binom{n-1}{k-1}}$$

ways to place the $k-1$ dividers and therefore distribute the balls. We note that this represents a more constrained and restricted form of the original problem. As an *exercise*, one can readily prove that the number of ways to distribute n balls into k bins, where the ith bin must have at least a_i balls, is equal to

$$\binom{n-(a_1+a_2+\cdots+a_k)+k-1}{k-1} = \binom{n-(\sum_{i=1}^{k} a_i)+k-1}{k-1}.$$

Note that we will have $n-(a_1+a_2+\cdots+a_k)$ free balls (balls not required to reside in any particular bin) and $k-1$ dividers.

APPLICATION

Exercise 1: When 7 fair standard 6-sided dice are thrown, the probability that the sum of the numbers on the top faces is 10 can be written as

$$\frac{n}{6^7}.$$

where n is a positive integer. What is n? (AMC 10A 2018)

Solution: We argue that 6^7 represents the total number of outcomes from 7 dice (6 outcomes for each dice), so n must represent the number of outcomes where the dice sum to 10. Thus, we need to find the number of solutions involving positive integers to the below equation, acknowledging that it is impossible for any $x_i > 6$ here,

$$x_1+x_2+x_3+x_4+x_5+x_6+x_7=10.$$

This is the equivalent to depositing 10 balls in 7 bins, with each bin having at least one ball, so we have $10-7=3$ balls that can be freely distributed with 6 dividers. Thus, the total number of distributions is $\binom{6+3}{3}=\binom{9}{3}=84$. We could alternatively use the formula $\binom{n-1}{k-1}=\binom{10-1}{7-1}=84$.

Figure 1: An example scenario with red balls as required placements and green balls as freely mobile balls that can moved from one bin to another.

Exercise 2: Pat is to select six cookies from a tray containing only chocolate chip, oatmeal, and peanut butter cookies. There are at least six of each of these three kinds of cookies on the tray. How many different assortments of six cookies can be selected? (AMC 10A 2003)

Solution: Let x_1, x_2, and x_3 be the quantity of chocolate chip, oatmeal, and peanut butter cookies, respectively. We aim to find the non-negative solutions to $x_1 + x_2 + x_3 = 6$ since we select six cookies, which is the same problem as separating a line of 6 cookies with 2 dividers. This can ensue in $\binom{6+2}{2} = \boxed{28}$ distinct ways, as this represents the number of ways to place 2 dividers among $6 + 2 = 8$ total slots.

Exercise 3: How many six-degree polynomials $f(x)$ with positive integer coefficients are there such that $f(1) = 30$ and $f(-1) = 12$? (AoPS)

Solution: We can represent the polynomial as shown below, with the aim of finding the number of possible tuples of positive coefficients.

$$f(x) = ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g.$$

We note that $f(1) = a+b+c+d+e+f+g = 30$ and $f(-1) = a-b+c-d+e-f+g = 12$. This means that

$$\begin{aligned}
f(1) + f(-1) &= a+b+c+d+e+f+g+(a-b+c-d+e-f+g) \\
&= 2a+2c+2e+2g \\
&= 42 \\
f(1) - f(-1) &= a+b+c+d+e+f+g-(a-b+c-d+e-f+g) \\
&= 2b+2d+2e \\
&= 18.
\end{aligned}$$

Essentially, we have $a+c+e+g = 21$ and $b+d+e = 9$. Using the balls and urns analogy, we note that there are $\binom{21-1}{4-1} = \binom{20}{3}$ positive integer solutions to the first equation and $\binom{9-1}{3-1} = \binom{8}{2}$ positive integer solutions to the second equation. Since no two variables are shared between the two equations, the total number of ways to select all coefficients is

$$\binom{20}{3} * \binom{8}{2} = \boxed{31920}. \tag{1}$$

Exercise 4: A gardener plants 3 maple trees, 4 oak trees, and 5 birch trees in a row. He plants them in random order, each arrangement being equally likely. Find the probability that no 2 birch trees are next to each other. (AIME 1984)

Solution 1 (Distributions): Let B and N be birch and non-birch trees, respectively. We recognize that the row must have the below structure since birch trees cannot be adjacent.

$$_B_B_B_B_B_$$

Note that the middle four slots must have at least one N to separate the birch trees from one another, while the leftmost and rightmost slots are unregulated. The number of ways to distribute the 7 Ns among 6 slots is the same as solving the following equation for positive integers:

$$x_1 + (x_2 - 1) + (x_3 - 1) + (x_4 - 1) + (x_5 - 1) + x_6 = 7 - 4$$
$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 3.$$

There are $\binom{3+6-1}{6-1} = 56$ distributions. For any particular distribution, such as $NNBNBNBNBNBN$, we have $\binom{7}{3} = 35$ ways to selecting where the 3 maple trees will reside among the 7 Ns; the oak trees will occupy the rest. Hence, the number of tree arrangements with non-adjacent birch trees is $56 * 35$.

To figure out the total number of tree arrangements, we consider a row of 12 slots with one tree per slot. We first place the 5 birch trees, which can occur in $\binom{12}{5}$ ways, and then place the 4 oak trees in the remaining 7 slots, which can occur in $\binom{7}{4}$ ways. The maple trees will reside in the remaining slots, so the number of unrestricted arrangements is

$$\binom{12}{5} * \binom{7}{4} = 27720.$$

Thus, the desired probability is $\frac{56*35}{27720} = \boxed{\frac{7}{99}}$.

Solution 2: Let B and N symbolize birch and non-birch trees, respectively. We recognize that the row must have the following structure since birch trees cannot be adjacent.

$$_N_N_N_N_N_N_N_$$

Here, we have 8 slots to place 5 birch trees, which can occur in $\binom{8}{5} = 56$ ways. The total quantity of tree arrangements with no regulations is equal to $\binom{3+4+5}{5} = \binom{12}{5}$, since there are 12 total places in the row and we need to place the 5 birch trees, with the remaining 7 places

reserved for non-birch trees. Note that it does not matter contextually which of the Ns are oak or maple trees. Accordingly, the probability that no birch trees are adjacent is

$$\frac{\binom{8}{5}}{\binom{12}{5}} = \frac{\frac{8!}{5!3!}}{\frac{12!}{5!7!}} = \frac{8!7!}{12!3!} = \boxed{\frac{7}{99}}.$$

PROBLEMS A

1. In how many ways can 20 peanut butter bars be distributed to 5, assuming that each student must receive at least two peanut butter bars?

2. In the 2014 minigame Castle Siege from MinePlex, each player on the Defending Team can select one of the following roles: Marksman, Knight, and Paladin. If there are 40 total players on the Defending Team, and there must be at least 5 Marksmen, 10 Knights, and 5 Paladin, how many possible distributions of roles are there? A sample distribution would be 10 Marksmen, 15 Knights, and 15 Paladin.

3. Find the number of ordered quintuples $(x_1, x_2, x_3, x_4, x_5)$ of positive odd integers that satisfy the following:
$$x_1 + x_2 + x_3 + x_4 + x_5 = 50$$
Hint: Note that a positive odd integer x_i can be written as $2k_i - 1$ for $k_i \geq 1$.

4. In a town election involving 100 voters, each voter can cast a ballot for one of 4 candidates: Joe, Bernie, Elizabeth, or Kamala. What is the number of possible distributions of votes among the 4 candidates? A sample distribution would be 50 for Joe, 15 for Elizabeth, 15 for Bernie, and 20 for Kamala. It is possible for a candidate to receive anywhere from 0 to 100 votes.

5. For some particular value of N, when $(a + b + c + d + 1)^N$ is expanded and like terms are combined, the resulting expression contains exactly 1001 terms that include all four variables a, b, c, and d, each to some positive power. (AMC 10A 2016)

   (a) Note that the exponents of any term $a^{x_a}b^{x_b}c^{x_c}d^{x_d}1^{x_e}$ in this expansion must sum to N. Equivalently,
   $$x_a + x_b + x_c + x_d + x_e = N$$
   Note that each of x_a, x_b, x_c, x_d must be positive, while x_e has more flexibility and must be non-negative (0 is possible), as elaborated in the problem.

   Here, we have 5 bins and $N - 4$ freely movable balls, since 4 of the balls (one for each of x_a, x_b, x_c, x_d) are deposited ahead of time. How many ways are there to deposit $N - 4$ balls in 5 bins without restriction?

   (b) Setting the expression obtained in (a) equal to 1001, what is N? This fully solves the designated AMC 10 problem.

6. Suppose that a subway train in the game *Subway Surfers* features a row of 20 adjacent seats stretching from one end of the train to the other. Due to the need for social distancing, no two people can be in adjacent seats. How many ways are there to seat 5 people in the row? *Consider the arrangement* _P_P_P_P_P_ *with* 6 *slots and* 5 *people. How many ways can we distribute unoccupied seats into the slots so that no two people are sitting adjacently?*

7. (Challenge) Prove the hockey-stick identity, which states that

$$\binom{k}{k} + \binom{k+1}{k} + \cdots + \binom{n}{k} = \sum_{i=k}^{n} \binom{i}{k} = \binom{n+1}{k+1}.$$

Hint: Note that $\binom{n+1}{k+1}$ *represents the number of ways to deposit* $n+2$ *balls in* $k+2$ *urns with the baseline requirement of at least one ball per urn. In general,* $\binom{i}{k}$ *represents the number of ways to deposit* $i+1$ *balls in* $k+1$ *urns. To make the connection, peform casework based on the number of balls deposited in the* $(k+2)$*th urn and count the number of ways to deposit the remaining balls.*

PROBLEMS B

1. Alice has 24 apples. In how many ways can she share them with Becky and Chris so that each of the three people has at least two apples? (AMC 8 2019)

2. Ed has five identical green marbles, and a large supply of identical red marbles. He arranges the green marbles and some of the red ones in a row and finds that the number of marbles whose right hand neighbor is the same color as themselves is equal to the number of marbles whose right hand neighbor is the other color. An example of such an arrangement is GGRRRGGRG. Let m be the maximum number of red marbles for which such an arrangement is possible, and let N be the number of ways he can arrange the $m+5$ marbles to satisfy the requirement. Find the remainder when N is divided by 1000. (AIME II 2011)

3. In a certain lottery, 7 balls are drawn at random from n balls numbered 1 through n. If the probability that no pair of consecutive numbers is drawn equals the probability of drawing exactly one pair of consecutive numbers, find n. (Mandelbrot)

4. Let $D(n)$ denote the number of ways of writing the positive integer n as a product

$$n = f_1 \cdot f_2 \cdots f_k,$$

where $k \geq 1$, the f_i are integers strictly greater than 1, and the order in which the factors are listed matters (that is, two representations that differ only in the order of the factors are counted as distinct). For example, the number 6 can be written as 6, $2 \cdot 3$, and $3 \cdot 2$, so $D(6) = 3$. What is $D(96)$? (AMC 10B 2020)

SOLUTIONS

Exercise: In how many ways can 20 peanut butter bars be distributed to 5, assuming that each student must receive at least two peanut butter bars?

Solution: We first distribute the mandatory two bars to the 5 people, which leaves us with 10 remaining candies. Thus, we have 10 candies and 5-1=4 dividers, which can arranged in $\binom{10+4}{10} = \boxed{1001}$ ways.

Exercise: In the 2014 minigame Castle Siege from MinePlex, each player on the Defending Team can select one of the following roles: Marksman, Knight, and Paladin. If there are 40 total players on the Defending Team, and there must be at least 5 Marksmen, 10 Knights, and 5 Paladin, how many possible distributions of roles are there? A sample distribution would be 10 Marksmen, 15 Knights, and 15 Paladin.

Solution: We first assign the necessary roles, which include 5 Marksmen, 10 Knights, and 5 Paladin, leaving behind $40 - 5 - 10 - 5 = 20$ assortable roles. We can imagine solving the equation $x + y + z = 20$, where x, y, and z describe the additional number of Marskmen, Knights, and Paladin, respectively. This is the same as having 20 balls and 3-1 dividers, which can arranged in $\binom{20+2}{2} = \boxed{190}$ ways.

Exercise: Find the number of ordered quintuples $(x_1, x_2, x_3, x_4, x_5)$ of positive odd integers that satisfy the following:

$$x_1 + x_2 + x_3 + x_4 + x_5 = 50$$

Hint: Note that a positive odd integer x_i can be written as $2k_i - 1$ for $k_i \geq 1$.

Solution: We note that a positive odd integer x_i can be written as $2k_i - 1$ for $k_i \geq 1$. Thus, we can write:

$$\begin{aligned}(2k_1 - 1) + (2k_2 - 1) + (2k_3 - 1) + (2k_4 - 1) &= 50\\ 2(k_1 + k_2 + k_3 + k_4) &= 54\\ k_1 + k_2 + k_3 + k_4 &= 27.\end{aligned}$$

We note that k_1, k_2, k_3, k_4 must each be positive, meaning that the number of solutions is $\binom{27-1}{4-1} = \boxed{2600}$.

Exercise: For some particular value of N, when $(a+b+c+d+1)^N$ is expanded and like terms are combined, the resulting expression contains exactly 1001 terms that include all four variables a, b, c, and d, each to some positive power. (AMC 10A 2016)

Solution: Note that the exponents of any term $a^{x_a}b^{x_b}c^{x_c}d^{x_d}1^{x_e}$ in this expansion must sum to N. Equivalently,

$$x_a + x_b + x_c + x_d + x_e = N$$

We imagine having N balls to be deposited in 5 bins. Note that each of x_a, x_b, x_c, x_d must be positive, while x_e has more flexibility and can be 0. Thus, we assign four balls first, leaving us with $N-4$ balls to be freely distributed. With $N-4$ balls and 4 dividers, we have $\binom{N}{4}$ possible arrangements. Setting this equal to 1001 (total number of terms), we get $\boxed{N=14}$.

Exercise: Suppose that a subway train in the game *Subway Surfers* features a row of 20 adjacent seats stretching from one end of the train to the other. Due to the need for social distancing, no two people can be in adjacent seats. How many ways are there to seat 5 people in the row?

Solution: We consider the arrangement $_P_P_P_P_P_$ with 6 slots and 5 people, noting that the middle four slots need to be filled with at least one seat for social distancing. Let x_i be the number of seats deposited into slot i for $1 \leq i \leq 6$. Thus, we have:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 15.$$

This is because there are $20 - 5 = 15$ unoccupied seats. We first deposit four seats, one into each of the middle four slots, to socially distance the passengers, leaving behind 11 seats to freely distribute. This means 11 seats with $6-1$ dividers, which can arranged in $\binom{16}{11} = \boxed{4368}$ ways.

Exercise: (Challenge) Prove the hockey-stick identity, which states that

$$\binom{k}{k} + \binom{k+1}{k} + \cdots + \binom{n}{k} = \sum_{i=k}^{n} \binom{i}{k} = \binom{n+1}{k+1}.$$

Solution: Note that $\binom{n+1}{k+1}$ represents the number of ways to deposit $n+2$ balls in $k+2$ urns with the baseline requirement of at least one ball per urn.

We can solve the same problem along a different strategy. We perform casework based on the number of balls deposited in the $(k+2)$th urn and count the number of ways to deposit the remaining balls.

We can deposit anywhere between 1 and $n+2-(k+1)$ balls in the $(k+2)$th urn (each urn must get at least one ball), which means that inversely anywhere between $k+1$ and $n+1$ balls can be distributed elsewhere. The number of ways to distribute $k+1$ balls into $k+1$ remaining bins is $\binom{k}{k}$, $k+2$ balls into $k+1$ remaining bins is $\binom{k+1}{k}$, and onward, all the way up to distributing $n+1$ balls into $k+1$ bins in $\binom{n}{k}$ ways. Thus, summing up all of these cases leads to

$$\binom{k}{k} + \binom{k+1}{k} + \cdots + \binom{n}{k}.$$

Since both $\binom{k}{k} + \binom{k+1}{k} + \cdots + \binom{n}{k}$ and $\binom{n+1}{k+1}$ solve the same problem of distributing $n+2$ balls into $k+2$ urns with at least a ball per urn, this must be equal.

Exercise: In a certain lottery, 7 balls are drawn at random from n balls numbered 1 through n. If the probability that no pair of consecutive numbers is drawn equals the probability of drawing exactly one pair of consecutive numbers, find n. (Mandelbrot)

Solution: we first count the number of ways to draw no pair of consecutive numbers. We envision that the 7 balls will serve as dividers for the other $n-7$ balls, partitioning them into 8 groups. We can place these dividers either between them or on the left or right sides, giving us $n-6$ slots to place 7 balls, which can occur in $\binom{n-6}{7}$ ways.

We proceed to count the other case as well. First, the consecutive two balls can be placed in any of $n-6$ slots. The other drawn five balls can be placed in any of $n-7$ slots between balls (one slot is eliminated due to the consecutive balls). Thus, we can lottery draw the balls in $(n-6) * \binom{n-7}{5}$ ways.

We set the two expressions equal to one another.

$$(n-6) * \binom{n-7}{5} = \binom{n-6}{7}$$
$$\boxed{n = 54}.$$

Exercise: Call a positive integer *monotonous* if it is a one-digit number or its digits, when read from left to right, form either a strictly increasing or a strictly decreasing sequence. For example, 3, 23578, and 987620 are monotonous, but 88, 7434, and 23557 are not. How many monotonous positive integers are there?

Solution: We bifurcate our task into counting the number of strictly increasing and strictly decreasing numbers. We start with the former. We have 9 possible digit choices: 1 through 9. We note that there are $\binom{9}{i}$ ways of forming a strictly increasing number with i digits, since once we select i digits, there is only one way of arranging them in strictly increasing fashion. Thus, the total quantity of these numbers is

$$\binom{9}{1} + \binom{9}{2} + \cdots + \binom{9}{9} = 2^9 - \binom{9}{0} = 511.$$

The equality is due to the binomial theorem identity. We then calculate the number of strictly decreasing numbers, where we can use 10 digit choices: 0 through 9. There are $\binom{10}{i}$ ways of forming a strictly increasing number with i digits, since once we select i digits from 10, there is only one way of arranging them in strictly increasing fashion. The only exception is 0, which is not positive, so we can correct by subtracting 1.

$$\binom{10}{1} + \binom{10}{2} + \cdots + \binom{10}{10} - 1 = 2^{10} - \binom{10}{0} - 1 = 1022.$$

Thus, we add the two numbers to produce $511 + 1022 = 1533$, but we need to subtract the numbers 1 through 9 due to overcounting (1 through 9 are both strictly increasing and decreasing), so the final answer is $1533 - 9 = \boxed{1524}$.

The aforementioned binomial identity, in particular, is written below.

$$(1+1)^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n$$

Exercise: Let N denote the number of 7 digit positive integers have the property that their digits are in increasing order. Determine the remainder obtained when N is divided by 1000. (Repeated digits are allowed.) (AoPS Mock AIME)

Solution: For a 7-digit integer, let x_i be the number of times that the digit i appears. Since the digits are increasing and thus cannot be 0, then $1 \leq i \leq 9$. We note that, as there are 7 total digits,

$$\sum x_i = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 = 7.$$

For any distribution of digits, there is only one way to assemble them in increasing order. This is essentially placing 7 balls in 9 bins, so the number of distributions (7 balls arranged with $9 - 1 = 8$ dividers) is equal to $\binom{7+8}{7} = \binom{15}{7} = 6435$, which leaves a remainder of $\boxed{435}$ when divided by 1000.